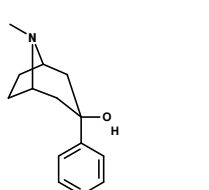
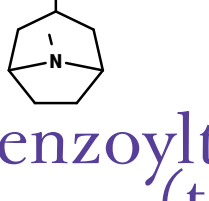
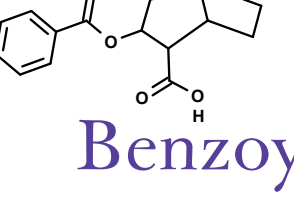
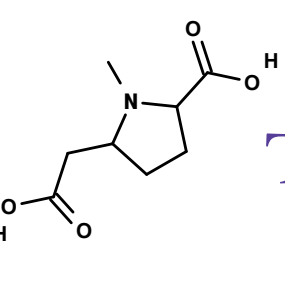
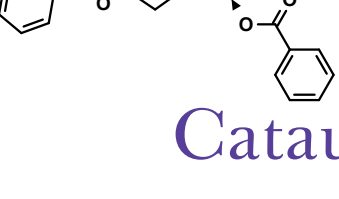
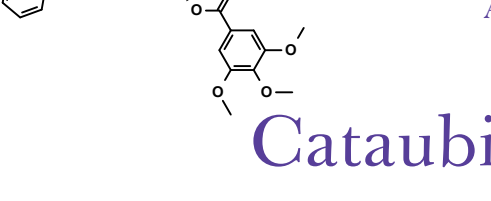
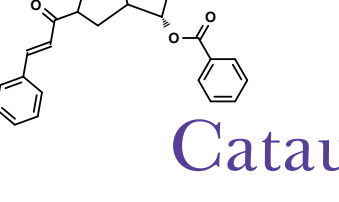
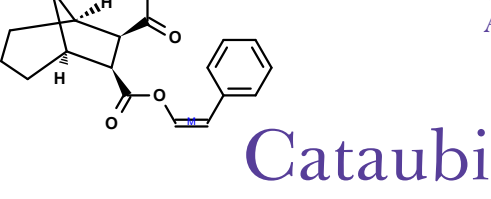
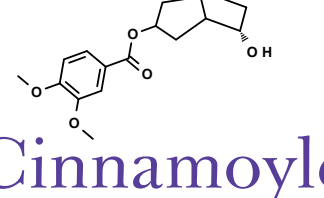
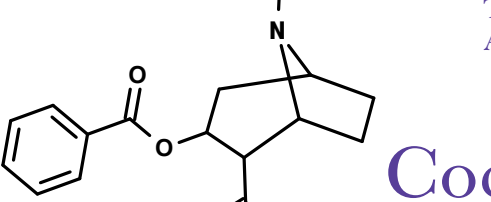
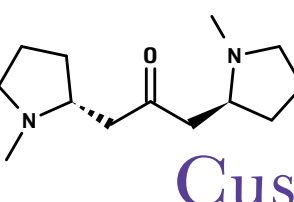
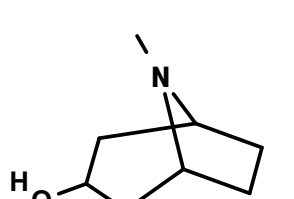
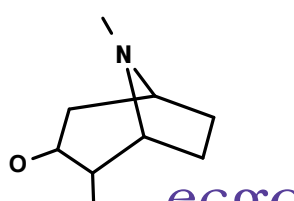
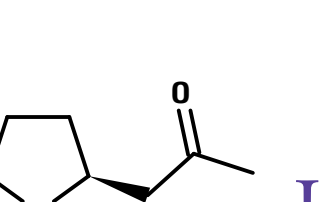
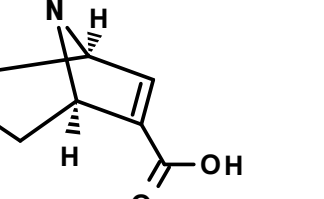
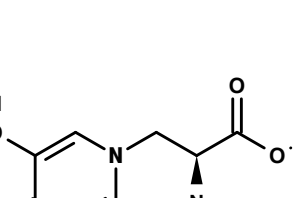
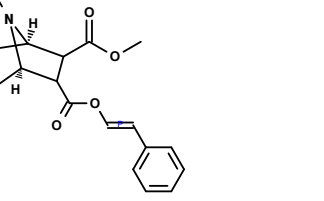
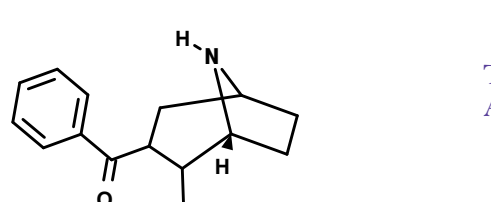
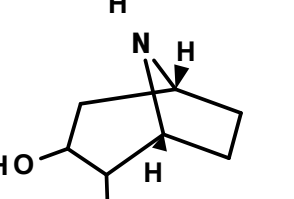
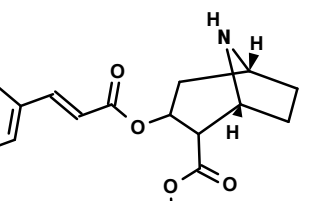


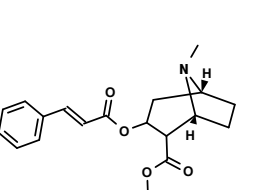
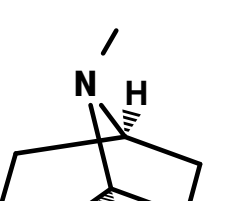
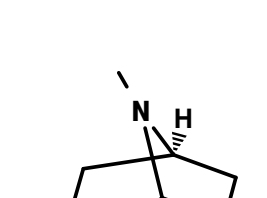
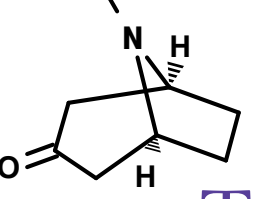
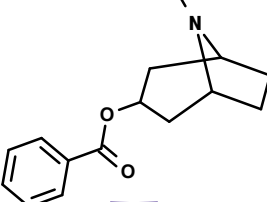


MAP OF SECONDARY METABOLITES OF COCA

Alkaloids

 3a-Benzoylhydroxy-tropane TROPANIC ALKALOID Irritant	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	+	-	-	-	-	-	Evans (1981)
 3-α-Benzoyltropane (tropine) TROPANIC ALKALOID Not reported	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-	-	-	Evans (1981)
 Benzoylecgonine acid TROPANIC ALKALOID Analgesic, anti-inflammatory	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	0.70%	1.87%	4.11%	1.84%			Findlay (1954)
 Tropine acid TROPANIC ALKALOID Antispasmodic effects	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	0.44%	0%	0.28%	0.07%			Jowett (1909)
 Benzoylecgonine TROPANIC ALKALOID Local anesthetic, stimulant effects	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	+	+	+	+			Evans (1981)
 Cataubine A TROPANIC ALKALOID Antimicrobial, antiproliferative, hemolytic	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Oliviero (2012)
 Cataubine B TROPANIC ALKALOID Antimicrobial, antiproliferative, hemolytic	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Oliviero (2012)
 Cataubine C TROPANIC ALKALOID Antimicrobial, antiproliferative, hemolytic	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Oliviero (2012)
 Cataubine D TROPANIC ALKALOID Antimicrobial, antiproliferative, hemolytic	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Oliviero (2012)
 Cis-Cinnamoylcocaine TROPANIC ALKALOID Leaf protection (possible)	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	+	+	+	+			Plewman (1983), Cosale (2006)
 Cocaine TROPANIC ALKALOID Local anesthetic, stimulant effects	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	3.13%	12.21%	4.43%	4.43%			Goedke, F. (1855), Niemann, A. (1860)
 Cuscohygrine PYRROLIDINE ALKALOID Antihypertensive	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Evans (1981), Ohtsuno, (2016)
 Ecgonine TROPANIC ALKALOID Cocaine precursor, effects on fatigue	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	11.21%	3.43%	9.93%	6.04%			Niemann, A. (1860)
 Ethyl ecgonine ester TROPANIC ALKALOID Analgesic and relaxing effects	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	2.60%	0%	0.53%	1.79%			Horns (1963)
 Hygrine PYRROLIDINE ALKALOID Forensic toxicology uses	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Evans (1981)
 Methyl ecgonidine TROPANIC ALKALOID Stimulant and analgesic effects	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Evans (1981)
 Mimosine PYRROLIDINE ALKALOID Potential antitumor, low toxicity	REFERENCES	Khoka project							REFERENCES
	(-)	(+)	0.01%	-	-	-			Adams (1945)
 Methyl ecgonine ester TROPANIC ALKALOID Analgesic, possible effect on the nervous system	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	5.71%	1.19%	-	-			Plewman (1983)
 Norcocaine TROPANIC ALKALOID Stimulant and analgesic effects	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Yulian (2022)
 Norecgonine TROPANIC ALKALOID Stimulant and analgesic effects	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Evans (1981)
 Trans-Cinnamoylecgonine TROPANIC ALKALOID Leaf protection	REFERENCES	Khoka project							REFERENCES
	(-)	(-)	-	-	-	-			Plewman (1983)

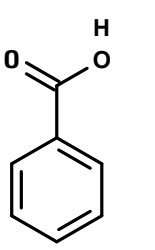
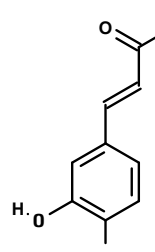
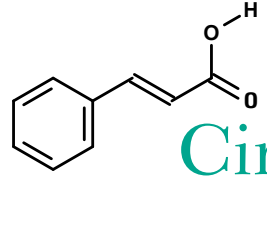
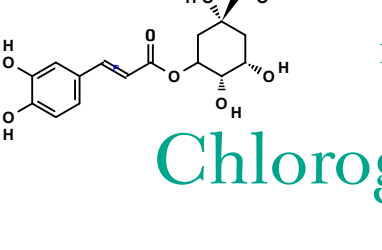
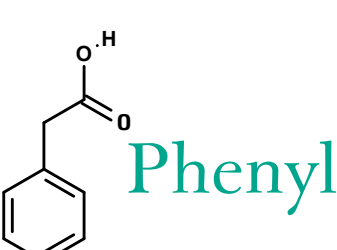
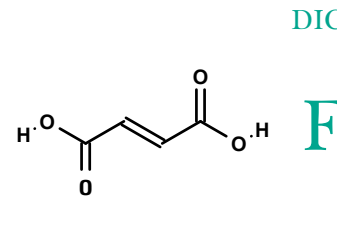
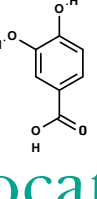
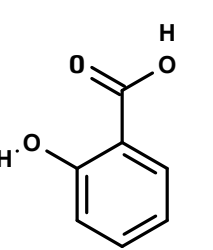
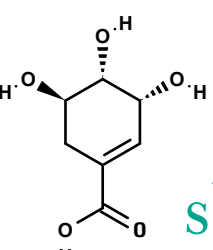
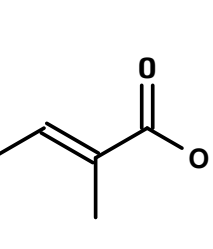
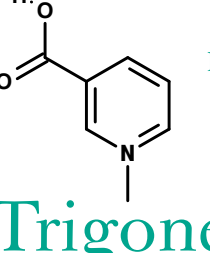
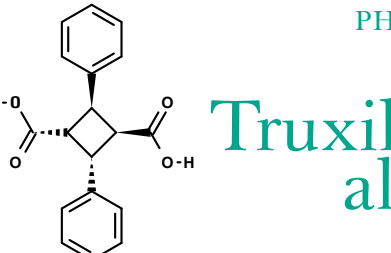
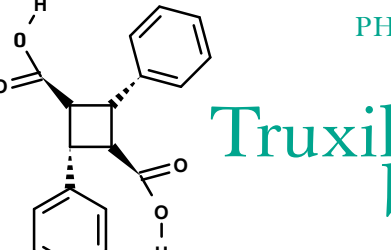
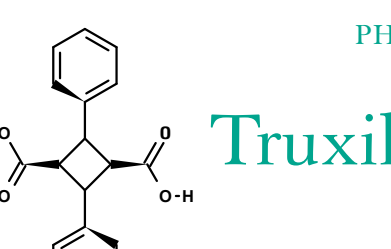
Alkaloids

 Trans-Cinnamoyl cocaine TROPANIC ALKALOID Possible anti-inflammatory effects	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	0.20%	0.44%	1.71%	2.24%			Plewman (1983)
 Tropane TROPANIC ALKALOID Stimulant and analgesic effects	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Yulian (2022)
 Tropine TROPANIC ALKALOID Stimulant and analgesic effects	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Yulian (2022)
 Tropinone TROPANIC ALKALOID Stimulant and analgesic effects	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Cheng (1981)
 Tropacocaine TROPANIC ALKALOID Stimulant and analgesic effects	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Jowett (1909)

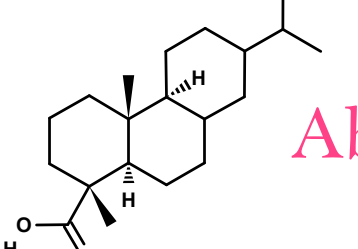
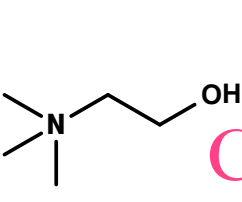
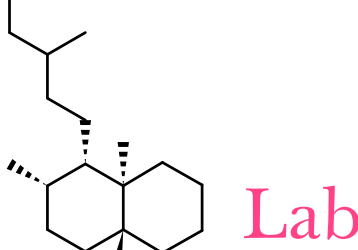
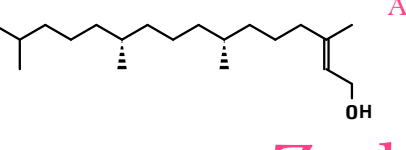
Acids

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Acids

 Benzoic acid BENZOIC ACID Preservative	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	+	+	+	+			Evans (1981)
 Caffeic acid PHENOLIC ACID Antioxidant, anti-inflammatory	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	+	0.04%	0.08%	-			Evans (1981)
 Cinnamic acid PHENOLIC ACID Antioxidant	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	+	0.12%	+	0.52%			Evans (1981)
 Chlorogenic acid PHENOLIC ACID Antioxidant, possible antidiabetic effect	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	0.34%	0.11%	+	-			Rivier (1981)
 Phenylacetic acid PHENOLIC ACID Preservative	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	+	+	+	+			Evans (1981)
 Ferulic acid DICARBOXYLIC ACID Preservative	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	0.04%	0.05%	0.15%	-			Evans (1981)
 Protocatechuic acid PHENOLIC ACID Antioxidant, anti-inflammatory	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	0.04%	0.03%	0.04%	0.05%			Yulian (2022)
 Salicylic acid PHENOLIC ACID Anti-inflammatory, analgesic	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	0.18%	0.05%	0.75%	0.41%			Hamel(1975)
 Ácido shikímico PHENOLIC ACID Antiviral drug precursor	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	0.16%	0.38%	1.08%	0.2%			Evans(1993)
 Tiglic acid UNSATURATED ACID Flavoring	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	+	+	+	+			Evans (1981)
 Trigonelline PHENOLIC ACID Neuroprotective, memory enhancement	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	0.10%	0.04%	0.05%	0%			Kuroda (2018)
 Truxilline alpha PHENOLIC ACID Protection against ruminants	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Evans (1981)
 Truxilline beta PHENOLIC ACID Protection against ruminants	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Evans (1981)
 Truxilline PHENOLIC ACID Protection against ruminants	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Evans (1981)

Others

 Abietic acid TERPENES Anti-inflammatory, healing	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	3.04%	6.33%	1.54%	2.31%			Evans (1981)
 Choline QUATERNARY AMINE Nutrient	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Cosale (2006)
 Labdane TERPENE Repellent	REFERENCES	Khoka project							REFERENCES
	(+)	(-)	-	-	-	-			Yulian (2022)
 Z-phytol ALCOHOL Repellent	REFERENCES	Khoka project							REFERENCES
	(+)	(+)	1.08%	0.19%	1.12%	1.22%			Ramirez (2018)